

ISM-fuzzy MICMAC approach for modelling the enablers of sustainability

Nida and Ashish Chandra

Department of Commerce, University of Delhi, New Delhi, India, and

Ruchi Shukla

Vivekananda Institute of Professional Studies, New Delhi, India

Abstract

Purpose – The objectives of this study are as follows: RO1: to determine the key enablers that encourage the adoption of sustainable practices; RO2: to create a structural relationship model between the enablers that have been discovered and RO3: to use dependence and driving power to classify and analyse these factors.

Design/methodology/approach – In total, 11 variables were extracted through systematic literature review and experts opinion, and a group interview with more than 15 experts who have an experience of more than 10 years in academics was conducted. Generally odd in numbers, they presented each pair of variables and took their opinion. Using “Interpretative Structural Modelling” and “Cross-Impact Multiplication Applied to Classification” (ISM MICMAC) analysis, a model describing the structural relationship between these factors has been formulated to know the conceptual framework.

Findings – The study shows that the digital payment and waste treatment as the linkage or mediating variables. Top management commitment (TMC), regulatory framework and transparency are the drivers or independent variables. Green logistics, sustainable packaging and production are the dependent variables. Further, the resulting hierarchy and contextual linkages among variables shed light on key dimensions for decision-making. Green logistics, sustainable packaging and production are interconnected, playing a significant role, in the framework and influenced by the lower level variables. Transparency should be given due consideration to ensure sustainable competitive advantage.

Research limitations/implications – (1) Owing to a lack of time and resources, the researchers have been able to explore only limited variables; more factors or enablers can be taken in future for further research. (2) Money constraint is also there. (3) One major limitation is that this research includes opinion from different experts which are dynamic in nature and keep on changing from time to time. (4) Primary research can also be done with the help of different software like partial least squares and structural equation modelling (PLS-SEM), covariance-based structural equation modelling (CB-SEM) and Statistical Package for the Social Sciences (SPSS) by taking same variables or merging new variables with the given variables of this paper. (5) With the help of this paper, the author tried to give an overview of some important enablers which are helpful in sustainability. Given a time in future, more research can be done by taking different sectors or a specific sector.

Practical implications – Findings are helpful for policymakers about effective strategies for promoting sustainability. They provide evidence-based insights into the potential impact of policy interventions and help shape the development of environmental regulations and initiatives. Sustainability is a global issue. Thus, this research paper contributes to international discussions and collaborations on environmental conservation and sustainable development goals (SDGs). They may influence diplomatic efforts to address transnational environmental challenges. Companies can use research findings to improve their sustainability practices and develop environmentally friendly products and services. Understanding sustainability enablers can help businesses reduce their ecological footprint while remaining economically viable. Research on sustainability enablers often involves exploring new technologies and innovations. Practical implications include the development and adoption of renewable energy sources, waste reduction techniques and sustainable



agricultural practices. Sustainable communities rely on a range of enablers, from renewable energy infrastructure to local governance structures. Research informs community development efforts by identifying best practices and effective strategies for building resilient, environmentally friendly neighbourhood.

Social implications – This research paper contributes to raising awareness about sustainability issues among the public. It also provides insights into the importance of sustainable practices and their impacts on society. Sustainability enablers can influence individual and collective behaviour, and they may encourage people to adopt more sustainable lifestyles and consumption habits. Sustainability research often addresses issues of equity and social justice. Understanding the enablers of sustainability can help identify strategies to ensure that sustainable practices benefit all segments of society, including marginalized communities.

Originality/value – This research paper endeavour to identify key sustainability development enablers (SDEs) by consolidating numerous factors within a single study. While several research papers have explored various sustainability enablers separately, no previous study has delved into their interactions. Furthermore, there is a dearth of research on classifying enablers based on the fuzzy MICMAC analysis. The ISM-based model presented here serves as a valuable resource for both practitioners and academics, aiding in their comprehension of the framework of relationships.

Keywords Sustainability, Sustainability development, Sustainability enablers, ISM, Fuzzy, MICMAC

Paper type Research paper

1. Introduction

The environment has suffered greatly as a result of human civilisation and industrialisation. Rapid urban development has resulted in the development of buildings, which have become a massive consumer of energy (Yuan, Wang, & Zuo, 2013). Because industries and other development projects have been carried out with little or no consideration for environmental protection, the situation has gotten worse (Geelani *et al.*, 2012). At this moment, we are grappling with numerous environmental challenges, which encompass a continuously broadening spectrum of pollutants, hazards and the degradation of ecosystems across increasingly extensive areas, which require long-term potential actions (Dincer, 2000). “Sustainability” is the central idea that revolves around integrating short-run and long-run problems” (Bhavsar & Samanta, 2021), and the ability to properly safeguard the environment, natural resources and the lives of future generations is known as sustainability (Keskin, Altay, Kurt, & Fleming, 2020). Physical, chemical, biological, social and economic elements all have an impact on the environment and are related to each community and population in different ways (Erfani, 2019).

The globally recognised aim of sustainable development within the development sector is to avoid the exhaustion of natural resources and ensure ecological equity for both present and future generations of humanity (Sharma, Nigrawal, & Baredar, 2020). A global issue that is receiving a lot of social and political attention is global warming. In recent decades, processes of urbanisation and globalisation have accelerated, increasing the consumption of natural resources and pollution in the environment, which has led to climate change (Corrêa, de Oliveira, Abdalla, & Fernandes, 2022). For today’s civilisation, the rapid decline in fossil fuel use and environmental contamination are serious concerns (Acharya *et al.*, 2020). As per the 2014 Intergovernmental Panel on Climate Change (IPCC), “the use of fossil fuels in electricity generation is one of the major contributors to greenhouse gas emission (GHG) worldwide”, and for that, de-carbonisation is required to reduce CO₂ and other GHG to make our environment sustainable.

First, it was concluded from the few earlier studies that they were more inclined either towards green supply chain management, sustainable energy or single sustainability factors. However, most of the studies failed to describe the enablers individually. Therefore, it was decided to take other enablers of sustainability with the existing one in a single study. Second, the researchers tried to fill the gap given in the literature (Ali, 2022; Dube & Gawande, 2016), and their studies exclusively focused on manufacturing and green supply management. It is

clear that the literature lacks understanding of the specific factors that encourage the adoption of sustainable practices. Therefore, the researchers have advocated for further investigation to gain a comprehensive understanding to innovate a model.

Third, the present study makes a valuable contribution to the existing literature. A further reason for studying sustainability enablers is that since 1987, there has been a significant increase in the prominence and integration of the sustainability development idea across a range of sectors (Goni, Shukor, Mukhtar, & Sahran, 2015), which led to an increase in research. An increase in research projects resulted from the growing importance of sustainability development, encouraging writers to conduct studies meant to yield significant findings in the context of environmental sustainability. This research paper analyses the factors and/or enablers for influencing sustainability by summarising the literature to understand the specific solution for sustainable practices. This study seeks to make a significant contribution to the ongoing conversation about creating a more sustainable future by summarising the factors into a single literature.

The remaining part of the paper continues as follows. The literature is highlighted in the second section. The research technique applied to this study is presented in the third section, which is followed by the results, analysis, discussion and implications. The last sections highlight the conclusion ending with the limitations and future scope.

2. Literature review

The Brundtland Report, a document which was published in 1987, articulates, as per definition, “sustainable development is a development that meets the needs of the present without compromising the ability of future generations to meet their own needs”. The United Nations further escalated the same issue in 1992. As per the United Nations Environment Programme UNEP(2011), “Since then, global society has achieved some significant successes, such as the reduction in ozone-depleting chemicals and the increased use of renewable energy sources, as well as in decoupling emissions and economic development”. Organisations globally have embraced sustainability strategies with the aim of improving their efficiency and gaining a competitive edge (Bateh, Horner, Broadbent, & Fish, 2014). There are many factors which can enhance sustainability. Although more recent research has started to concentrate on connecting personal environmentalisms with group processes (Anantharaman, 2018), more and more researchers are emphasising “sustainability” as a key goal (Lee *et al.*, 2019). Since the consumption of items is rising globally, we know that the resulting harm and environmental effects are also increasing significantly (Corrêa *et al.*, 2022). Companies can aim to target new segment by using a holistic approach to sustainability in order to improve the market share of sustainable products (Camilleri, Cricelli, Mauriello, & Strazzullo, 2023). Green processes are essential for developing trustworthy marketing practices (Burki & Ersoy, 2022).

Erfani (2019) says that preserving the environment is equal to preserving human lives and to preserve the environment. Sustainability has a major role. It represents a social obligation, ethical consideration or a significant component within the framework of strategic management. Moreover, it has been interconnected with the process of strategic decision-making (Bateh *et al.*, 2014).

Many papers are available on different sectors like green logistics. Kumar (2015), in his paper, explains that green logistics is a crucial component of supply chain management (SCM) and helps to enhance the transportation network. de Souza, Kerber, Bouzon, and Rodriguez (2022) talk about the performance evaluation of green logistics, which is a strategic factor in sustainability (Chunguang, Xiaojuan, Kexi, & Pan, 2008; Seroka-Stolka, 2014; Tüzün Rad & Gülmez, 2017). The sugar industry’s supply chain is crucial to the industry’s overall sustainability (Kumar, Agrawal, & Sharma, 2015). Another sector is sustainable buildings,

which explains that considering sustainability in constructing buildings is helpful in the current as well as future advancement (Liu *et al.*, 2022); it is an important sustainable development factor (Balaban & Puppim de Oliveira, 2017) for healthier cities (Geelani *et al.*, 2012; Sharma *et al.*, 2020). The third sector is green supply chain management (GSCM). Kitchen and Vel (2005) and Wang and Liu Zuoming (2019) explain how it is impacting the governance of the corporates and their commitments. Dube and Gawande (2016) talk about its enablers. Further, Morashti, An, and Jang (2022) say the inculcation of sustainable packaging in the supply chain is helpful. Manufacturing is also one of the sector found during the literature review, and the enablers were assessed. Ali (2022), Toke and Kalpande (2019) and Jain and Hudnurkar (2022) talk about some strategic changes towards more sustainable and eco-friendly products. Kumar and Bangwal (2023) attempted to evaluate measures for a sustainable supply chain in the Indian car sector.

Research on various sectors is available, but there aren't many research papers on compiling various sustainability development enablers (SDEs). Studies were conducted to understand the interactions between individual sectors, such as GSCMEs and green manufacturing enablers, but there aren't any on classifying sustainability enablers overall based on fuzzy cross-impact matrix multiplication applied to classification (MICMAC) analysis. Keeping these gaps in mind, objectives and goals were framed.

Consequently, the following are the main goals of this investigation:

RO1. To determine the key enablers that encourage the adoption of sustainable practices.

RO2. To create a structural relationship model between the enablers that have been discovered.

RO3. To use dependence and driving power to classify and analyse these factors.

To get the results for first objective, a "systematic literature review" was done with expert opinions, and only those academicians (experts) who have an experience of more than 10 years were interviewed. To establish the structural relationships for the second and third objectives, "Total Interpretive Structural Modelling" (TISM) and "Cross-Impact Multiplication Applied to Classification" (MICMAC) analysis techniques were applied to classify these factors according to driving and dependence power.

To fill the gaps and achieve the objectives, the below factors are taken (Table 1) after a detailed analysis with the help of literature, research and expert opinions.

2.1 Identification of key enablers:

2.1.1 *Digital payment.* Globally, it has been asserted that digital payments (digital financial service) facilitate expansion and expedite the attainment sustainability (Bhavsar & Samanta, 2021). Mobile money, digital payments and more recent digital payment systems are examples of digital financial services (Bhavsar & Samanta, 2022). The Government of India's "Digital India" initiative has led to an exponential growth in digital payments in recent years (Bhavsar & Samanta, 2022). India's electronic payment system holds great potential for future success (Shree, Pratap, Saroy, & Dhal, 2021); utilising m-payment has become more popular during the past ten years (Abdullah & Naved Khan, 2021).

2.1.2 *Online grocery.* E-grocery, developed over time (Aziz *et al.*, 2022; Droogenbroeck & Van Hove, 2021; Gatta, Marcucci, Maltese, Iannaccone, & Fan, 2021), is growing fast worldwide (Prabowo, Hindarwati, & Yuniarty, 2020). The number of articles increased yearly from 2000 to 2017 (Martín, Pagliara, & Román, 2019), but due to the pandemic, they have increased significantly (Aziz *et al.*, 2022). Compared with physical stores, e-grocery may be more environmentally friendly, but when the basket size decreases, it starts losing its environmental advantage (Siragusa & Tumino, 2022). There may be fewer dedicated

Table 1.
Sustainability factors **Source(s):** Authors' composition

Factors	Authors
Energy	Acharya <i>et al.</i> (2020), Ahmad and Tahar (2014), Aziz <i>et al.</i> (2022), Coman, Florescu, and Oancea (2020), Della Valle and Bertoldi (2022), Dey <i>et al.</i> (2022), Dincer (2000), Erfani (2019), Garba Danjumma Sani (2019), Hafez <i>et al.</i> (2023), Charles Rajesh Kumar and Majid (2020), Jha (2023), Lesic, De Bruin, Davis, Krishnamurti, and Azevedo (2018), Lv and Zhang (2023), Nie, Zhou, Lu, and Huang (2021), Owusu and Asumadu-Sarkodie (2016), Perea-Moreno (2021), Qazi <i>et al.</i> (2019) and Yuan <i>et al.</i> (2013)
Production	Corrêa <i>et al.</i> (2022), Glavič (2021), Kukreti, Painolli, and Rana (2021), Lee <i>et al.</i> (2019), Lukman <i>et al.</i> (2016), Machado <i>et al.</i> (2020), Trivedi and Sharma (2017), Wang <i>et al.</i> (2019) and Zarte, Pechmann, and Nunes (2022)
Consumer awareness	Buerke <i>et al.</i> (2017), Camilleri <i>et al.</i> (2023), Ishak and Zabil (2012), Kukreti <i>et al.</i> (2021), Makanyeza, Svotwa, and Jaiyeoba (2021), Nikolić <i>et al.</i> (2022), Stauropoulou, Sardanou, Malindretos, Evangelinos, and Nikolaou (2023) and Harshith Babu <i>et al.</i> (2020)
Transparency	Ball (2009), Bateh <i>et al.</i> (2014), Fraser and van der Ven (2022), Gupta <i>et al.</i> (2020), Montecchi <i>et al.</i> (2021), and Ofem <i>et al.</i> (2022)
Top management commitment	Burki and Ersoy (2022), Lee and Joo (2020), Siagian <i>et al.</i> (2022), Singh and Rahman (2021), Wang and Liu Zuoming (2019), Yusliza <i>et al.</i> (2019)
Digital payment	Abdullah and Naved Khan (2021), Bhavsar and Samanta (2021, 2022), and Shree <i>et al.</i> (2021)
Sustainable packaging	Escursell, Llorach-Massana, and Roncero (2021), Jain and Hudnurkar (2022), Keskin <i>et al.</i> (2020), Morashti <i>et al.</i> (2022), Oloyede and Lignou (2021), Siracusa and Rosa (2018) and Varžinskas and Markevičiūtė (2020)
E-Grocery	Aziz <i>et al.</i> (2022), Droogenbroeck and Van Hove (2021), Gatta <i>et al.</i> (2021), Martín <i>et al.</i> (2019), Mkansi, Eresia-Eke, and Emmanuel-Ebikake (2018), Prabowo <i>et al.</i> (2020), Siragusa and Tumino (2022)
Waste treatment	Amasuomo and Baird (2016), Dhote, Pradeep Ingole, and Chavhan (2012), Farooqi <i>et al.</i> (2021), Gunarathne, Ashiq, and Ginige (2018), Kesari <i>et al.</i> (2021), Kumar <i>et al.</i> (2017) and Sikosana, Sikhwivhilu, Moutloali and Madyira (2019)
Regulatory framework	Madawaki (2014), Mishra and Kumar (2023), Mitra (2010) and Turaga and Sugathan (2020)
Green logistics	Bedingfield (2020), Chunguang <i>et al.</i> (2008), de Souza <i>et al.</i> (2022), Demir, Syntetos, and Van Woensel (2022), Kitchen and Vel (2005), Kumar (2015), Li <i>et al.</i> (2021), Seroka-Stolka (2014) and Tüzün Rad and Gülmez (2017)

journeys if click-and-collect services are used more frequently (Droogenbroeck & Van Hove, 2021).

2.1.3 Top management commitment. Social, environmental and economic sustainability all contribute to corporate sustainability; however, many sustainable development goals (SDGs) have received less attention from firms (Singh & Rahman, 2021). There is a positive relationship between top management commitment (TMC) and corporate social responsibility (Yusliza *et al.*, 2019). Furthermore, TMC has a direct effect on green process innovation (Burki & Ersoy, 2022) and the environmental performance of companies (Lee & Joo, 2020). The need for businesses to harmonise their economic and environmental performance has grown significantly, and a crucial method for achieving this balance is the commitment of the top management team to endorse the company's green initiatives (Wang & Liu Zuoming, 2019). Indeed, TMC plays a pivotal role and yields various benefits in bolstering competitive advantage through the establishment of strategies and policies (Siagian, Tarigan, & Basana, 2022).

2.1.4 Regulatory framework. The supporting structure that directs and carries out a suggested path of legislation, regulation or action is known as the regulatory framework (Mishra & Kumar, 2023), and the application of the law is necessary to preserve and repair ecological integrity (Dernbach & Mintz, 2011). Despite the fact that pollution has one of the highest human costs in the world, India was ironically among the first countries in the world to incorporate environmental considerations into its legislation and policymaking process, starting in the early 1970s (Turaga & Sugathan, 2020). India has started implementing regulatory changes since they are now necessary given the shift in macroeconomic policy (Mitra, 2010).

2.1.5 Consumer awareness. Awareness about the environment has found its place in consumers' mindset. The average customer is more conscious of their product choices these days (Jain & Hudnurkar, 2022). As a result, on the path to sustainable growth, consumers are placed in charge (Buerke, Straatmann, Lin-Hi, & Müller, 2017). Awareness is prior to effective consumers' behaviour (Ishak & Zabil, 2012). Prospective customers will consider a product or service's "greenness" to be advantageous and will choose it appropriately (Harshith Babu, Kumar, & Venkatesh, 2020). Responsible consumption should comprise a societal as well as an individual dimension (Buerke *et al.*, 2017). Sustainable consumption has attracted the interest of both governments and society due to mounting apprehension about resource exhaustion and the escalating volumes of waste production (Camilleri *et al.*, 2023). Understanding the mutual relation between awareness and consumer behaviour is crucial (Nikolić, Paunović, Milovanović, Lozović, & Đurović, 2022). There is a remarkable rise, and the last few decades have seen a rise in environmental awareness, which has put more pressure on consumers to think about how their actions may affect the environment (Ansu-Mensah, 2021).

2.1.6 Transparency. The call for transparency is stronger than ever. Transparency encourages openness (Ball, 2009) and emerging novel forms of digitalised transparency (Gupta, Boas, & Oosterveer, 2020). Organisations that are transparent have made significant efforts to raise stakeholder scrutiny and improve operational accountability (Montecchi, Plangger, & West, 2021). The fast fashion industry is subject to growing calls for transparency (Fraser & van der Ven, 2022); for instance, fashion retailer H&M has taken the initiative to furnish comprehensive details on the sourcing, manufacturing and sustainability of every product available for sale through their online store labelled under "transparency layer". Still many firms are not aligning with domestic norms to increase transparency (Fraser & van der Ven, 2022), so there is a greater need to focus on this enabler. There is a need to uncover transparency (Ofem, Isong, & Lugayizi, 2022).

2.1.7 Waste treatment. India confronts significant environmental problems as a result of improper waste management and trash generation (Kumar *et al.*, 2017). Wastes are materials whose owners no longer have a need for (Amasuomo & Baird, 2016). Waste is available in solid, industrial, municipal and water forms. Solid wastes (agricultural, industrial, domestic, E-waste, etc.) pose an environmental threat, and their management is of great concern in the modern era. However, different approaches are available (Farooqi, Kareem, Rafi, & Ali, 2021). Management involves a process whereby wastes is collected, transported and disposed of (Amasuomo & Baird, 2016). By implementing efficient waste management practices, societies can significantly reduce the strain on natural resources, minimise the need for raw resource extraction, save energy consumption and reduce greenhouse gas emissions (GGE).

2.1.8 Green logistics. The terms "green logistics" and "sustainability" refer to methods and approaches used in SCM and logistics that put the long-term health of the environment and human society ahead of other considerations. Sustainable development needs green logistics, which enhances green economic activity (Li, Sohail, Majeed, & Ahmad, 2021), since environmental issues have garnered a lot of attention globally. As a result, logistics companies are searching for ways to make their operations more environmentally friendly

(Seroka-Stolka, 2014), and COVID-19 pandemic has further emphasised the urgency. Green logistics issues are gaining high attention, which are closely linked to sustainability (Kumar, 2015). The three logistics functions, namely transportation, warehousing and packaging, can employ methods to reduce the effects on the environment (Bedingfield, 2020), and even the green production system is linked to the implementation of green logistics (de Souza *et al.*, 2022).

2.1.9 Sustainable packaging. Sustainability in packaging is a critical aspect of environmentally responsible practices within various industries. It encompasses the adoption of measures to reduce the environmental impact of packaging materials and methods. For decades, there has been a growing interest from all stakeholders to make packaging more sustainable (Oloyede & Lignou, 2021). Achieving sustainability in packaging can be accomplished through the utilisation of recycled packaging materials as well as the adoption of reusable and biodegradable packaging materials. Recycled packaging materials may encompass items such as pulp, corrugated cardboard, cardboard, newsprint paper, aluminium and certain types of plastics (Keskin *et al.*, 2020). Paper-based products are preferred by consumers because they are inexpensive, lightweight, durable, renewable and easily degraded in nature (Keskin *et al.*, 2020).

2.1.10 Production. Sustainable production practices are characterised by a commitment to minimising environmental impacts, conserving natural resources and promoting social responsibility by incorporating eco-friendly technologies and processes. Sustainable development is said to depend on a number of factors, including sustainable consumption and production (Wang, Ghadimi, Lim, & Tseng, 2019). The Brundtland Report's release has enhanced production procedures for sustainability (Lukman, Glavič, Carpenter, & Virtič, 2016). A greater degree of digitalisation is transforming industrial production systems, resulting in intelligent, connected and decentralised production. Further as per the analysis of Machado, Winroth, and Ribeiro da Silva (2020), current research ("Sustainable manufacturing research is contributing to the development of the Industry 4.0 agenda") is in line with the objectives set forth by several national industrial initiatives.

2.1.11 Energy. Energy is the ability or capacity to do work (Dey, Sreenivasulu, Veerendra, Rao, & Babu, 2022), and there has been a growing emphasis on prioritising energy efficiency as a strategic approach to reduce carbon emissions, limit energy usage, improve building energy efficiency and reduce overall energy consumption in order to attain sustainable building practices (Hafez *et al.*, 2023). In recent decades, discussions surrounding the promotion of energy efficiency have been primarily influenced by the physical-technical-economic model. This model places significant emphasis on technologies and cost savings while downplaying the role of human behaviour, often considering it a less significant factor (Della Valle & Bertoldi, 2022). The environmental and long-term sustainability are influenced by factors such as energy consumption during manufacturing processes (Geelani *et al.*, 2012).

There are many factors which can enhance sustainability. A total of 50% of household carbon dioxide emissions originate from equipment such as air conditioners, refrigerators and televisions, contributing around 70% of the emissions (Lv & Zhang, 2023). The urgency of addressing climate change necessitates a rapid reduction in energy consumption. One approach to accomplishing this is by enhancing energy efficiency or implementing energy-saving measures (Della Valle & Bertoldi, 2022), and the other is the usage of renewable energy, which contributes only 26.5% (Qazi *et al.*, 2019); instead of relying on non-renewable sources, fossil fuels continue to contribute 73.5% worldwide, which takes tons of years to generate. Ahmad and Tahar (2014) and Owusu and Asumadu-Sarkodie (2016) approached returning to renewables, which is to be one of the most efficient and effective solutions. Dincer (2000) and Perea-Moreno (2021) propose that, at all levels, conserving energy, using it wisely and utilising energy sources effectively are all vital.

Customers who perceive energy use and savings more accurately could be better equipped to pinpoint the activities that result in the greatest energy savings (Lesic *et al.*, 2018). Simultaneously, businesses are anticipated to employ finite resources in a manner that ensures enough is preserved for the well-being of future generations, safeguarding their quality of life from being compromised by the present generation's consumption (Bateh *et al.*, 2014). Finding sustainable energy sources, particularly renewable energy sources, should be a top priority for societies that aspire to or are already working towards sustainable development (Dincer, 2000). The use of renewable energy in buildings aids in diminishing the reliance on conventional energy sources and contributes to the realisation of a low-carbon economy (Yuan *et al.*, 2013). Hence, it is crucial to evaluate the strategies, procedures and options for improving the viability of the concept (Sharma *et al.*, 2020).

3. Research methodology

3.1 Identification of variables

This study is a collection of primary and secondary databases from different sources. First, an extensive literature review with experts opinion (Ali, 2022; Gupta & Singh, 2022; Toke & Kalpande, 2019) is done to find out various variables effecting or creating sustainability, and the situation of the environment is getting worse day by day, and negative climate situations like pandemics and epidemics are the reasons why we should focus on such types of research. In total, 11 variables were extracted through systematic literature review and expert opinions. A group interview with more than 15 experts who have more than 10 years of academic experience was conducted. Generally odd in numbers, they presented each pair of variables and took their opinion. Using the "Interpretative Structural Modelling" and "Cross-Impact Multiplication Applied to Classification" (ISM-MICMAC) analysis, a structural relationship between these factors has been formulated to know the conceptual framework (Dube & Gawande, 2016). The variables have been categorised as driver or dependent variables, which helped the researchers to understand the key variables. The complete procedure followed in this paper is portrayed through below figure.

This method is a powerful tool to use in decision-making processes across various fields, including project management, marketing and strategic planning. Its applicability stems from its ability to analyse interdependencies among different factors or variables within a complex system. By employing this method, stakeholders can gain valuable insights into the causal relationships between various elements, helping them identify key drivers and dependencies that influence the system's behaviour. One compelling reason to utilise the IMS-MICMAC analysis is its capacity to uncover both direct and indirect influences among factors. Traditional methods often focus solely on direct relationships, overlooking the subtler, yet significant, indirect effects. However, the IMS-MICMAC analysis comprehensively maps out these intricate connections, offering a more holistic understanding of the system dynamics.

Moreover, the method aids in prioritising interventions or decision-making actions by distinguishing between driving and dependent factors. This distinction enables stakeholders to allocate resources more efficiently, targeting interventions at the most influential variables to achieve desired outcomes effectively. Additionally, IMS-MICMAC's versatility makes it adaptable to various contexts and scales, whether analysing market dynamics, assessing risks in project management or formulating strategic plans. Its systematic approach and visual representation facilitate communication among stakeholders, fostering consensus-building and informed decision-making.

In essence, the ISM-MICMAC analysis serves as a valuable analytical tool for unravelling complexity, and the steps followed to identify key leverage points and guide effective decision-making in diverse domains, ultimately enhancing organisational performance and resilience in an ever-evolving environment, are represented by Figures 1 and 2

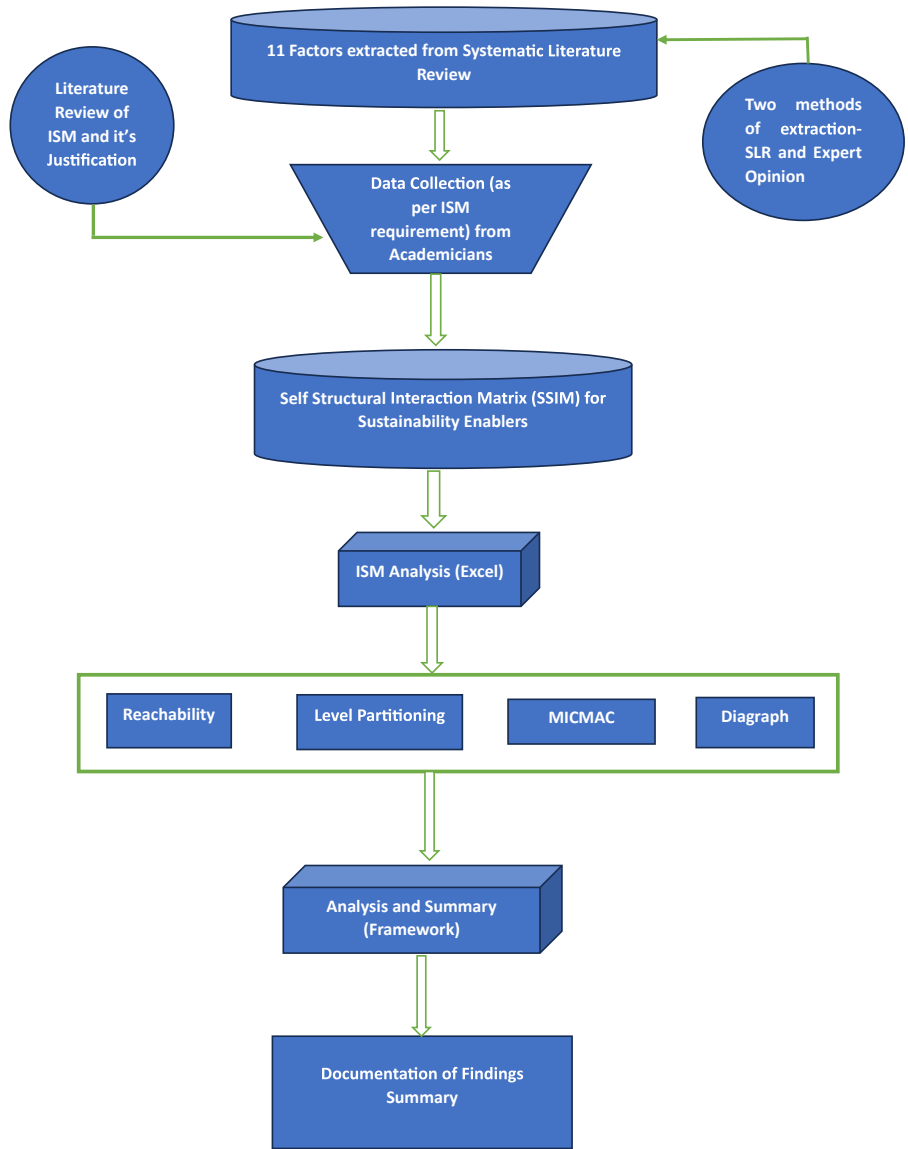


Figure 1.
Flow chart of the
process adopted for
interpretive structural
modelling

Source(s): Authors' composition

4. Results and analysis

4.1 Structural self-interaction matrix

After finalising the factors, the next step is to develop a conceptual framework to find out the relationship among them. This was done by pairing of factors in self-structural interaction matrix (SSIM), with the help of expert opinion. Four symbols were employed to represent the relationship between two variables, signifying both its direction and nature.

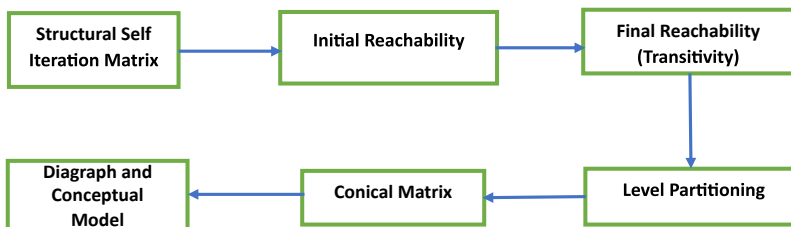
- (1) V – Variable I will help in achieving variable J;
- (2) A – Variable J will help in achieving variable I;
- (3) X –Both will lead to each other and
- (4) O- Variables are unrelated.

In Table 2, the symbols V, A, X and O are utilised also known as VAXO. Here, Variable 1 (DP*) leads to Variable 15 (E*), denoted with V; both Variable 2 (EG*) and Variable 15 (E*) are leading to each other, denoted with X, and similarly Variable A is showing that J variables leading to I variable and O in the matrix shows that there is no relationship between these two variables.

4.2 Reachability matrix

Subsequently, the SSIM was converted into a binary matrix, known as the initial reachability matrix (Table 3), through the replacement of V, A, X and O (Table 2) with the respective values of 1 and 0, depending on the circumstances.

- (1) Mark V as 1 when i Leading j and 0 in the opposite cell
- (2) Mark A as 0 when i leading j and 1 in the opposite cell
- (3) Mark X as 1 irrespective of direction



Source(s): Authors' composition

Figure 2.
Steps followed for ISM

ij	E*	PN*	SP*	GL*	WT*	T*	CA*	RF*	TMC*	EG*	DP*
DP*	V	O	O	V	X	A	X	X	A	X	–
EG*	X	V	V	V	V	A	V	A	A	–	
TMC*	V	V	V	V	V	A	A	A	–		
RF*	V	V	V	V	V	V	V	–			
CA*	V	V	V	V	V	X	–				
T*	O	V	V	V	V	–					
WT*	X	A	A	V	–						
GL*	A	V	V	–							
SP*	A	X	–								
PN*	A	–									
E*	–										

Note(s): DP - Digital payment, EG - E-Grocery, TMC - Top management Commitment, RF - Regulatory framework, CA - Consumer Awareness, T - Transparency, WT - Waste Treatment, GL - Green Logistics, SP - Sustainable packaging, PN - Production and E - Energy

Source(s): Authors' composition

Table 2.
Self-structural
interactive
matrix (SSIM)

Table 3.
Initial reachability
matrix

ij	DP	EG	TMC	RF	CA	T	WT	GL	SP	PN	E	Driving power
DP	1	1	0	1	1	0	1	1	0	0	1	7
EG	1	1	0	0	1	0	1	1	1	1	1	8
TMC	1	1	1	0	0	0	1	1	1	1	1	8
RF	1	1	1	1	1	1	1	1	1	1	1	11
CA	1	0	1	0	1	1	1	1	1	1	1	9
T	1	1	1	0	1	1	1	1	1	1	0	9
WT	1	0	0	0	0	0	1	1	0	0	1	4
GL	0	0	0	0	0	0	0	1	1	1	0	3
SP	0	0	0	0	0	0	1	0	1	1	0	3
PN	0	0	0	0	0	0	1	0	1	1	0	3
E	0	1	0	0	0	0	1	1	1	1	1	6
Dependence power	7	6	4	2	5	3	10	9	9	9	7	71

Source(s): Authors' composition

(4) Mark O as 0 irrespective of direction of the variables

After converting the matrix into binary table, reverse the row (Table 2) by putting Variable 15 as Variable 1 and so on and find the driving power (row wise) by adding all the values of particular variable; likewise, find the dependence power (column wise) by adding all the values of a particular variable.

Heading to the final reachability matrix (transitivity) in [Table 4](#), where we replace 0'S with 1 by using if $A = B, B=C$ then $A = C$, we will be getting a new matrix with different dependence and driving power values.

4.3 Level partitions

From the final reachability matrix, the reachability set and antecedent set were found (Table 5). The reachability set (row) includes the variables with a value of 1. Similarly, the antecedent set (column) consists of those variables, which have a value of 1, and ignores variables with zero values. Then, the intersection of these sets is derived. Those variables that have the same set in reachability as well as in intersection set like V8, V9 and V10 will be

ij	DP (v1)	EG (v2)	TMC (v3)	RF (v4)	CA (v5)	T (v6)	WT (v7)	GL (v8)	SP (v9)	PN (v10)	E (v11)	Driving power
DP (v1)	1	1	1*	1	1	1*	1	1	1*	1*	1	11
EG (v2)	1	1	1*	1*	1	1*	1	1	1	1	1	11
TMC (v3)	1	1	1	1*	1*	0	1	1	1	1	1	10
RF (v4)	1	1	1	1	1	1	1	1	1	1	1	11
CA (v5)	1	1*	1	1*	1	1	1	1	1	1	1	11
T (v6)	1	1	1	1*	1	1	1	1	1	1	1*	11
WT (v7)	1	1*	0	1*	1*	0	1	1	1*	1*	1	9
GL (v8)	0	0	0	0	0	0	1*	1	1	1	0	4
SP (v9)	1*	0	0	0	0	0	1	1*	1	1	1*	6
PN (v10)	1*	0	0	0	0	0	1	1*	1	1	1*	6
E (v11)	1*	1	0	0	1*	0	1	1	1	1	1	8
Dependence power	10	8	6	7	8	5	11	11	11	11	10	98

Table 4.
Final reachability
matrix (transitivity)

IJ	Reachability set A (row)	Antecedent set B (column)	Intersection set A∩B	Level
<i>First iteration</i>				
V1	1,2,3,4,5,6,7,8,9,10,11	1,2,3,4,5,6,7,9,10,11	1,2,3,4,5,6,7,9,10,11	
V2	1,2,3,4,5,6,7,8,9,10,11	1,2,3,4,5,6,7,11	1,2,3,4,5,6,7,11	
V3	1,2,3,4,5,7,8,9,10,11	1,2,3,4,5,6	1,2,3,4,5	
V4	1,2,3,4,5,6,7,8,9,10,11	1,2,3,4,5,6,7	1,2,3,4,5,6	
V5	1,2,3,4,5,6,7,8,9,10,11	1,2,3,4,5,6,7,11	1,2,3,4,5,6,7,11	
V6	1,2,3,4,5,6,7,8,9,10,11	1,2,4,5,6	1,2,4,5,6	
V7	1,2,4,5,7,8,9,10,11	1,2,3,4,5,6,7,8,9,10,11	1,2,4,5,7,8,9,10,11	
V8	7,8,9,10	1,2,3,4,5,6,7,8,9,10,11	7,8,9,10	1
V9	1,7,8,9,10,11	1,2,3,4,5,6,7,8,9,10,11	1,7,8,9,10,11	1
V10	1,7,8,9,10,11	1,2,3,4,5,6,7,8,9,10,11	1,7,8,9,10,11	1
V11	1,2,5,7,8,9,10,11	1,2,3,4,5,6,7,9,10,11	1,2,5,7,9,10,11	
<i>Second iteration</i>				
V1	2,3,4,5,6	2,3,4,5,6	2,3,4,5,6	2
V2	2,3,4,5,6	2,3,4,5,6,7,11	2,3,4,5,6	2
V3	2,3,4,5	2,3,4,5,6	2,3,4,5	2
V4	2,3,4,5,6	2,3,4,5,6	2,3,4,5,6	2
V5	2,3,4,5,6	2,3,4,5,6	2,3,4,5,6	2
V6	2,3,4,5,6	2,4,5,6	2,4,5,6	
V7	2,4,5	2,3,4,5,6	2,4,5	2
V11	2,5	2,3,4,5,6,7,9,10,11	2,5	2
<i>Third iteration</i>				
V6	2,3,4,5,6	2,4,5,6	2,4,5,6	3
<i>Final reachability matrix</i>				
Digital payment (V1)	1,2,3,4,5,6,7,8,9,10,11	1,2,3,4,5,6,7,9,10,11	1,2,3,4,5,6,7,9,10,11	2
E-grocery (V2)	1,2,3,4,5,6,7,8,9,10,11	1,2,3,4,5,6,7,11	1,2,3,4,5,6,7,11	2
Top management commitment (V3)	1,2,3,4,5,7,8,9,10,11	1,2,3,4,5,6	1,2,3,4,5	2
Regulatory framework (V4)	1,2,3,4,5,6,7,8,9,10,11	1,2,3,4,5,6,7	1,2,3,4,5,6	2
Consumer awareness (V5)	1,2,3,4,5,6,7,8,9,10,11	1,2,3,4,5,6,7,11	1,2,3,4,5,6,7,11	2
Transparency (V6)	1,2,3,4,5,6,7,8,9,10,11	1,2,4,5,6	1,2,4,5,6	3
Waste treatment (V7)	1,2,4,5,7,8,9,10,11	1,2,3,4,5,6,7,8,9,10,11	1,2,4,5,7,8,9,10,11	2
Green logistics (V8)	7,8,9,10	1,2,3,4,5,6,7,8,9,10,11	7,8,9,10	1
Sustainable packaging (V9)	1,7,8,9,10,11	1,2,3,4,5,6,7,8,9,10,11	1,7,8,9,10,11	1
Production (V10)	1,7,8,9,10,11	1,2,3,4,5,6,7,8,9,10,11	1,7,8,9,10,11	1
Energy (V11)	1,2,5,7,8,9,10,11	1,2,3,4,5,6,7,9,10,11	1,2,5,7,9,10,11	2

Source(s): Authors' composition

Table 5.
Partitioning of
reachability matrix

given the top level in the hierarchy. Start the second iteration (Table 5) of the matrix after deleting the previous variable and intersecting variables, and the process will go on with the third iteration (Table 5) till we get the final matrix (Table 5) with all variables with their set level, as shown in the tables below.

4.4 Building the isometric model

Following the segmentation of the levels, connections among different factors are established using a diagram. This entails creating a node for each variable and linking them with arrows

according to the direction of their relationships. Subsequently, the diagram is initially assessed and confirmed for transitivity, as described in the methodology, before it is transformed into an ISM model.

Figure 3, shows some very interesting results.

- (1) Green logistics, sustainable packaging and production are at the primary or first level, basically emerging as the starting point and the most relevant factors of the model. They are interconnected and impacting each other, which shows they are working together for sustainability.
- (2) Transparency is at the third and/or final level, which means it is the least recognised factor and impacts the variables at second level, which are also impacting each other simultaneously.
- (3) Variables at the second level are digital payment, E-grocery, TMC and regulatory framework, which impact variables at the level 1, besides impacting each other.

After the diagram is checked and confirmed to be transitive – a process that is elucidated in the methodology – it is transformed into an ISM model.

4.5 MICMAC analysis

MICMAC is employed to assess the driving power and dependence power of the variables, which serves as the basis for their categorisation into four distinct groups, which can be seen in Figure 4:

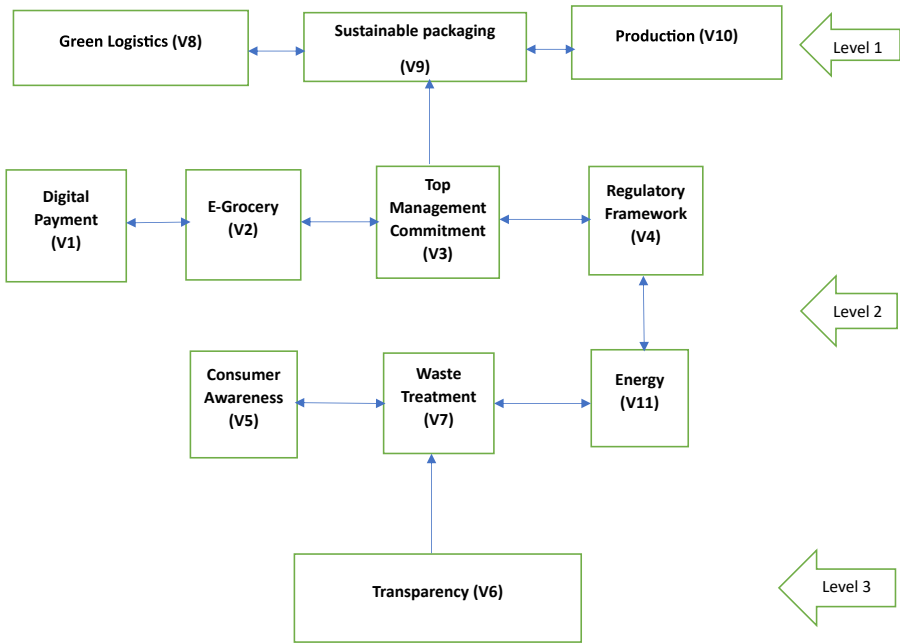


Figure 3.
Isometric model

Source(s): Authors' composition

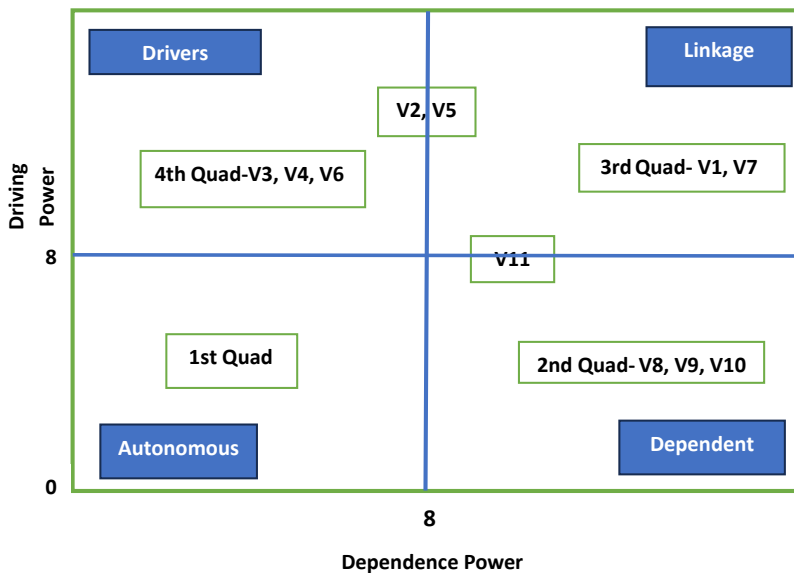


Figure 4.
MICMAC matrix

Source(s): Authors' composition

- (1) Autonomous (1st Quadrant) - They are also known as isolated variables; if any variable comes in this quadrant, it should be removed because they are low driving as well as dependent power.
- (2) Dependent (2nd Quadrant) - V8, V9 and V10 (green logistics, sustainable packaging and production) are the three variables lying in this quadrant, showing purely dependent variables, which can also be seen in the isometric model.
- (3) Linkage (3rd Quadrant) - V1, V7 (digital payment and waste treatment) are lying in this quadrant, showing mediating variables.
- (4) Driving variables (4th Quadrant), V3, V4 and V6 (TMC, regulatory framework and transparency) are the variables belong to this quadrant, showing purely independent variables.

With the help of [Table 6](#), [Figure 4](#) was made.

V1 (10,11)
V2 (8,11)
V3 (6,10)
V4 (7,11)
V5 (8,11)
V6 (5,11)
V7 (11,9)
V8 (11,4)
V9 (11,6)
V10 (11,6)
V11 (10,8)

Source(s): Authors' composition

Table 6.
MICMAC matrix –
cluster of variables

5. Discussion

5.1 *Novel contribution of this paper*

- (1) This research paper endeavours to identify key SDEs by consolidating numerous factors within a single study.
- (2) While several research papers have explored various sustainability enablers, separately, no previous study has delved into their interactions.
- (3) Furthermore, there is a dearth of research on classifying enablers based on the fuzzy MICMAC analysis.
- (4) The ISM-based model presented here serves as a valuable resource for both practitioners and academics, aiding in their comprehension of the framework of relationships.

A key finding of this research is that green logistics, sustainable packaging and production are significant SDEs. These are at the bottom right corner of the MICMAC analysis and implying dependent variable. Management should be more focused on transparency, which is at the bottom of the conceptual framework.

It was found that sustainability factors or enablers implementation have become essential. Sustainability factors play a crucial role in shaping the present and future of our planet. Though many organisations benefited, it is challenging and difficult to implement. At its core, sustainability encompasses the responsible consumption of resources and the preservation of ecological balance to meet the needs of current and future generations. Embracing sustainability is essential for several reasons. Firstly, it addresses the pressing issue of climate change by reducing carbon emissions and promoting renewable energy sources. Secondly, sustainability fosters economic resilience by encouraging innovation and efficiency in resource utilisation, leading to long-term cost savings and increased competitiveness. Thirdly, it safeguards biodiversity and ecosystems, ensuring the continuity of essential ecosystem services such as clean air, water and fertile soil. Additionally, sustainability promotes social equity by considering the needs of marginalised communities and promoting fair labour practices. Ultimately, integrating sustainability factors into decision-making processes across sectors is crucial for building a more resilient, equitable and thriving world for current and future generations.

6. Implications of the paper

Research papers on sustainability and its enablers can have significant social, political and practical implications:

6.1 *Social Implications:*

- (1) This research paper contributes to raising awareness about sustainability issues among the public. It also provides insights into the importance of sustainable practices and their impacts on society.
- (2) Sustainability enablers can influence individual and collective behaviour, and they may encourage people to adopt more sustainable lifestyles and consumption habits.
- (3) Sustainability research often addresses issues of equity and social justice.
- (4) Understanding the enablers of sustainability can help identify strategies to ensure that sustainable practices benefit all segments of society, including marginalised communities.

6.2 Political Implications:

- (1) Findings are helpful for policymakers about effective strategies for promoting sustainability. They provide evidence-based insights into the potential impact of policy interventions and help shape the development of environmental regulations and initiatives.
- (2) Sustainability is a global issue. Thus, this research paper contributes to international discussions and collaborations on environmental conservation and the SDGs. They may influence diplomatic efforts to address transnational environmental challenges.

6.3 Practical implications:

- (1) Companies can use research findings to improve their sustainability practices and develop environmentally friendly products and services. Understanding sustainability enablers can help businesses reduce their ecological footprint while remaining economically viable.
- (2) Research on sustainability enablers often involves exploring new technologies and innovations. Practical implications include the development and adoption of renewable energy sources, waste reduction techniques and sustainable agricultural practices.
- (3) Sustainable communities rely on a range of enablers, from renewable energy infrastructure to local governance structures. Research informs community development efforts by identifying best practices and effective strategies for building a resilient, environmentally friendly neighbourhood.

Consumers are more aware today and inclining to purchase eco-friendly products. E-commerce and E-retailing companies can inculcate such factors into their business models to create more awareness among consumers. Consequently, they can benefited from aligning their profit motives with social responsibilities.

Overall, this research paper on sustainability and its enablers plays a crucial role in shaping social attitudes, informing policy decisions and driving practical actions towards a more sustainable future.

7. Conclusion

India is committed to the SDGs of the United Nations (Singh & Rahman, 2021). Sustainable development is the theme of the future (Chunguang *et al.*, 2008). Commitment to the natural environment has become an important variable (Seroka-Stolka, 2014). An increase in research projects has resulted from the growing importance of sustainability development, encouraging writers to conduct studies meant to yield significant findings in the context of environmental sustainability. This research paper analyses the factors influencing sustainability by summarising the literature. The object of this paper is to identify the enablers to implement sustainability and to understand their mutual relationship. Few studies in the past have studied the variables using the ISM-MICMAC analysis (Dube & Gawande, 2016) and identified 14 key variables of GSCM, according to which its implementation is tough task due to the complexity of GSCM practices, customer awareness, cost of product and regulation. Singh, Panackal and Sharma (2016) tried to identify and modelling of environmental awareness and accounting affecting industry and sustainability with five variables. With eight variables, came up the main driving forces for successful implementation of green manufacturing, which are “economic constraints” and “regulatory framework”, as

discussed by (Ali, 2022). In total, 16 inhibitors were identified by Khan *et al.* (2020) with the help of literature and brainstorming method to evaluate transparency in humanitarian logistics. Gupta and Singh (2022) studied 11 barriers in the Indian digital payment sector. Further, to fill gaps in the given research article, 11 variables were extracted through systematic literature review and expert opinions. Using “Interpretative Structural Modelling” and “Cross-Impact Multiplication Applied to Classification” (ISM-MICMAC) analysis, structural relationships among these factors have been developed to know the conceptual framework (which is the proposed relationship between the variables). The factors have been classified as driver or dependent variables. The study reveals the digital payment and waste treatment as the linkage or mediating variables, TMC (Toke & Kalpande, 2019), regulatory framework and transparency as the drivers or independent variables and green logistics, sustainable packaging and production are the dependent variables. Further, the resulting hierarchy and contextual linkages among variables shed light on key dimensions for decision-making, green logistics, sustainable packaging and production are interconnected, playing a significant role in the framework and influenced by the lower level variables. The second level variables are digital payment, E-grocery, TMC, regulatory framework, energy, waste treatment and consumer awareness are interconnected to each other, which are influenced by transparency and influencing top-level variables also. “Transparency”, which is at the last level, should be given due consideration to ensure sustainable competitive advantage. Transparency builds trust among stakeholders and allows for informed decision-making, enabling businesses and governments to align their actions with sustainable goals. Consumer awareness, waste treatment and energy, which are at second level also plays a major, consumer awareness empowers individuals to make environmentally conscious choices, driving demand for sustainable products and services and influencing market dynamics. Effective waste treatment mechanisms mitigate environmental pollution and resource depletion while promoting circular economy principles. Additionally, prioritising energy efficiency and transitioning to renewable sources reduces reliance on finite resources and mitigates climate change impacts. By focusing on these areas, we not only address immediate environmental challenges but also lay the foundation for a more resilient and equitable future. Embracing transparency, enhancing consumer awareness, improving waste treatment and optimising energy management are imperative steps towards achieving sustainability goals and securing a flourishing planet for generations to come.

The outcomes of this research are likely to be valuable for both industry practitioners and academics. This is because it encompasses a wide range of factors while also taking into account both the internal and external aspects of a firm. Additionally, the paper pinpoints gaps in existing literature and provides insights into potential avenues for future research.

8. Limitations and future scope

There are some limitations too, which need to be focused:

- (1) Owing to a lack of time and resources, the researchers have been able to explore only limited variables. More factors or enablers can be taken in future for further research.
- (2) Money constraint is also there.
- (3) One major limitation is that this research includes opinion from different experts which are dynamic in nature and keep on changing from time to time.
- (4) Primary research can also be done with the help of different software like partial least squares and structural equation modelling (PLS-SEM), covariance-based structural equation modelling (CB-SEM) and Statistical Package for the Social Sciences (SPSS) by taking same variables or merging new variables with the given variables of this paper.

- (5) With the help of this paper, the authors tried to give an overview of some important enablers, which are helpful in sustainability. Given a time in future, more research can be done by taking different sectors or a specific sector.

A key finding of this research is that green logistic, sustainable packaging and production are significant SDEs. Simultaneously, these are at the lower most right corner of the MICMAC analysis, implying dependent variable. Management should be more focused on transparency which is at the bottom of conceptual framework.

References

- Abdullah, & Naved Khan, M. (2021). Determining mobile payment adoption: A systematic literature search and bibliometric analysis. *Cogent Business and Management*, 8(1). doi: [10.1080/23311975.2021.1893245](https://doi.org/10.1080/23311975.2021.1893245).
- Acharya, S., Shil, A., Debbarma, C., Reang, J., Chakraborty, R., & Ghosh, A. (2020). Analysis of energy consumption, emission and saving opportunities in an educational institute in northeast India. *International Journal of Energy and Water Resources*, 4(4), 375–388. doi: [10.1007/s42108-020-00086-1](https://doi.org/10.1007/s42108-020-00086-1).
- Ahmad, S., & Tahar, R. M. (2014). Selection of renewable energy sources for sustainable development of electricity generation system using analytic hierarchy process: A case of Malaysia. *Renewable Energy*, 63, 458–466. doi: [10.1016/j.renene.2013.10.001](https://doi.org/10.1016/j.renene.2013.10.001).
- Ali, S. S. (2022). Green manufacturing: An assessment of enablers' framework using ISM-MICMAC analysis. *Foundations of Computing and Decision Sciences*, 47(3), 271–290. doi: [10.2478/fcds-2022-0015](https://doi.org/10.2478/fcds-2022-0015).
- Amasuomo, E., & Baird, J. (2016). The concept of waste and waste management. *Journal of Management and Sustainability*, 6(4), 88. doi: [10.5539/jms.v6n4p88](https://doi.org/10.5539/jms.v6n4p88).
- Anantharaman, M. (2018). Critical sustainable consumption: A research agenda. *Journal of Environmental Studies and Sciences*, 8(4), 553–561. doi: [10.1007/s13412-018-0487-4](https://doi.org/10.1007/s13412-018-0487-4).
- Ansu-Mensah, P. (2021). Green product awareness effect on green purchase intentions of university students': An emerging market's perspective. *Future Business Journal*, 7(1), 1–13. doi: [10.1186/s43093-021-00094-5](https://doi.org/10.1186/s43093-021-00094-5).
- Aziz, S., Maltese, I., Marcucci, E., Gatta, V., Benmoussa, R., & Irhirane, E. H. (2022). Energy consumption and environmental impact of E-grocery: A systematic literature review. *Energies*, 15(19), 7289. doi: [10.3390/en15197289](https://doi.org/10.3390/en15197289).
- Balaban, O., & Puppim de Oliveira, J. A. (2017). Sustainable buildings for healthier cities: Assessing the co-benefits of green buildings in Japan. *Journal of Cleaner Production*, 163, S68–S78. doi: [10.1016/j.jclepro.2016.01.086](https://doi.org/10.1016/j.jclepro.2016.01.086).
- Ball, C. (2009). What is transparency?. *Public Integrity*, 11(4), 293–308. doi: [10.2753/PIN1099-9922110400](https://doi.org/10.2753/PIN1099-9922110400).
- Bateh, J., Horner, D. H. J., Broadbent, A., & Fish, D. (2014). Sustainability: A literature review and suggested way forward. *Journal of Sustainability Management*, 2(1), 35–42. doi: [10.19030/jsm.v2i1.8756](https://doi.org/10.19030/jsm.v2i1.8756).
- Bedingfield, D., & Abcam (2020). Environmental sustainability in logistics. 1149099036. Available from: <https://discover.abcam.com/News/21628/environmental-sustainability-in-logistics>
- Bhavsar, V., & Samanta, P. K. (2021). Investigating the sustainability of digital payments in India. *Orissa Journal of Commerce*, 42(2), 95–108. doi: [10.54063/ojc.2021.v42i02.08](https://doi.org/10.54063/ojc.2021.v42i02.08).
- Bhavsar, V., & Samanta, P. K. (2022). Sustainability of digital payments : Empirical evidence from India. 2, 70–78.
- Buerke, A., Straatmann, T., Lin-Hi, N., & Müller, K. (2017). Consumer awareness and sustainability-focused value orientation as motivating factors of responsible consumer behavior. *Review of Managerial Science*, 11(4), 959–991. doi: [10.1007/s11846-016-0211-2](https://doi.org/10.1007/s11846-016-0211-2).

- Burki, U., & Ersoy, P. (2022). Top management pledge, an essential component of sustainable manufacturer-customer relationships. *Journal of Sustainable Marketing*, 3(2), 98–117. doi: [10.51300/jsm-2022-61](https://doi.org/10.51300/jsm-2022-61).
- Camilleri, M. A., Cricelli, L., Mauriello, R., & Strazzullo, S. (2023). Consumer perceptions of sustainable products: A systematic literature review. *Sustainability (Switzerland)*, 15(11), 8923. doi: [10.3390/su15118923](https://doi.org/10.3390/su15118923).
- Charles Rajesh Kumar, J., & Majid, M. A. (2020). Renewable energy for sustainable development in India: Current status, future prospects, challenges, employment, and investment opportunities. *Energy, Sustainability and Society*, 10(1), 1–36. doi: [10.1186/s13705-019-0232-1](https://doi.org/10.1186/s13705-019-0232-1).
- Chunguang, Q., Xiaojuan, C., Kexi, W., & Pan, P. (2008). Research on green logistics and sustainable development. In *Proceedings of the International Conference on Information Management Management/Proceedings of the International Conference on Information Management* (Vol. 3, pp. 162–165). Innovation Management and Industrial Engineering, ICIII 2008. doi: [10.1109/ICIII.2008.114](https://doi.org/10.1109/ICIII.2008.114).
- Coman, C. M., Florescu, A., & Oancea, C. D. (2020). Improving the efficiency and sustainability of power systems using distributed power factor correction methods. *Sustainability (Switzerland)*, 12(8), 1–20. doi: [10.3390/SU12083134](https://doi.org/10.3390/SU12083134).
- Corrêa, R. D. S., de Oliveira, U. R., Abdalla, M. M., & Fernandes, V. A. (2022). Systematic literature review on sustainable products: Impact on organizations, research opportunities and future perspectives. *Cleaner Waste Systems*, 1(January), 100003. doi: [10.1016/j.clwas.2022.100003](https://doi.org/10.1016/j.clwas.2022.100003).
- de Souza, E. D., Kerber, J. C., Bouzon, M., & Rodriguez, C. M. T. (2022). Performance evaluation of green logistics: Paving the way towards circular economy. *Cleaner Logistics and Supply Chain*, 3(December 2021), 100019. doi: [10.1016/j.clscn.2021.100019](https://doi.org/10.1016/j.clscn.2021.100019).
- Della Valle, N., & Bertoldi, P. (2022). Promoting energy efficiency: Barriers, societal needs and policies. *Frontiers in Energy Research*, 9(February), 1–12. doi: [10.3389/ferg.2021.804091](https://doi.org/10.3389/ferg.2021.804091).
- Demir, E., Syntetos, A., & Van Woensel, T. (2022). Last mile logistics: Research trends and needs. *IMA Journal of Management Mathematics*, 33(4), 549–561. doi: [10.1093/imaman/dpac006](https://doi.org/10.1093/imaman/dpac006).
- Dernbach, J. C., & Mintz, J. A. (2011). Environmental laws and sustainability: An introduction. *Sustainability*, 3(3), 531–540. doi: [10.3390/su3030531](https://doi.org/10.3390/su3030531).
- Dey, S., Sreenivasulu, A., Veerendra, G. T. N., Rao, K. V., & Babu, P. S. S. A. (2022). Renewable energy present status and future potentials in India: An overview. *Innovation and Green Development*, 1(1), 100006. doi: [10.1016/j.igd.2022.100006](https://doi.org/10.1016/j.igd.2022.100006).
- Dhote, J., Pradeep Ingole, S., & Chavhan, A. (2012). (PDF) review on waste water treatment technologies. *Review on Wastewater Treatment Technologies*, 1(5), 1–18. Available from: https://www.researchgate.net/publication/230595234_Review_on_Waste_Water_Treatment_Technologies
- Dincer, I. (2000). Renewable energy and sustainable development: A crucial review. *Renewable and Sustainable Energy Reviews*, 4(2), 157–175. doi: [10.1016/S1364-0321\(99\)00011-8](https://doi.org/10.1016/S1364-0321(99)00011-8).
- Droogenbroeck, E. V., & Van Hove, L. (2021). Adoption and usage of E-grocery shopping: A context-specific UTAUT2 model. *Sustainability (Switzerland)*, 13(8), 4144. doi: [10.3390/su13084144](https://doi.org/10.3390/su13084144).
- Dube, A. S., & Gawande, R. R. (2016). ISM-fuzzy MICMAC approach for analysis of GSCM enablers. *International Journal of Logistics Systems and Management*, 24(4), 426–451. doi: [10.1504/IJLSM.2016.077281](https://doi.org/10.1504/IJLSM.2016.077281).
- Erfani, H. (2019). A case study on factors affecting environmental sustainability development A case study on factors affecting environmental sustainability development. December.
- Escursell, S., Llorach-Massana, P., & Roncero, M. B. (2021). Sustainability in e-commerce packaging: A review. *Journal of Cleaner Production*, 280, 124314. doi: [10.1016/j.jclepro.2020.124314](https://doi.org/10.1016/j.jclepro.2020.124314).
- Farooqi, Z. U. R., Kareem, A., Rafi, F., & Ali, S. (2021). Solid waste, treatment technologies, and environmental sustainability: Solid wastes and their sustainable management practices. In *Handbook of Research on Waste Diversion and Minimization Technologies for the Industrial Sector*, December (pp. 35–57). doi: [10.4018/978-1-7998-4921-6.ch003](https://doi.org/10.4018/978-1-7998-4921-6.ch003).

- Fraser, E., & van der Ven, H. (2022). Increasing transparency in global supply chains: The case of the fast fashion industry. *Sustainability (Switzerland)*, 14(18), 11520. doi: [10.3390/su141811520](https://doi.org/10.3390/su141811520).
- Garba, D. S. (2019). Renewable energy: Environmental impacts and economic benefits for sustainable development. *International Journal of Engineering Research*, V8(08), 547–555. doi: [10.17577/ijertv8is080224](https://doi.org/10.17577/ijertv8is080224).
- Gatta, V., Marcucci, E., Maltese, I., Iannaccone, G., & Fan, J. (2021). E-Groceries: A channel choice analysis in shanghai. *Sustainability (Switzerland)*, 13(7), 1–18. doi: [10.3390/su13073625](https://doi.org/10.3390/su13073625).
- Geelani, S. M., Geelani, S. H., Bhat, S. J. A., Haq, S., Mir, N. A., Junaid, S., & Zafar, B. (2012). Green building development for sustainable environment with special reference to India. *International Journal of Environment and Bioenergy J. Environ. Bioener*, 4(2), 86–100. Available from: www.ModernScientificPress.com/Journals/IJEE.aspx
- Glavič, P. (2021). Evolution and current challenges of sustainable consumption and production. *Sustainability (Switzerland)*, 13(16), 9379. doi: [10.3390/su13169379](https://doi.org/10.3390/su13169379).
- Goni, F. A., Shukor, S. A., Mukhtar, M., & Sahran, S. (2015). Environmental sustainability: Research growth and trends. *Advanced Science Letters*, 21(2), 192–195. doi: [10.1166/asl.2015.5850](https://doi.org/10.1166/asl.2015.5850).
- Gunarathne, V., Ashiq, A., & Ginige, M. P. (2018). Green adsorbents for pollutant removal. 18 (June 2018). doi: [10.1007/978-3-319-92111-2](https://doi.org/10.1007/978-3-319-92111-2).
- Gupta, A., & Singh, S. (2022). Barriers of digital transaction in rural areas: An interpretive structural modelling and MICMAC analysis. *International Journal of Electronic Business*, 18(1), 108–128. doi: [10.1504/ijeb.2023.127541](https://doi.org/10.1504/ijeb.2023.127541).
- Gupta, A., Boas, I., & Oosterveer, P. (2020). Transparency in global sustainability governance: To what effect?. *Journal of Environmental Policy and Planning*, 22(1), 84–97. doi: [10.1080/1523908X.2020.1709281](https://doi.org/10.1080/1523908X.2020.1709281).
- Hafez, F. S., Sa'di, B., Safa-Gamal, M., Taufiq-Yap, Y. H., Alrifay, M., Seyedmahmoudian, M., . . . Mekhilef, S. (2023). Energy efficiency in sustainable buildings: A systematic review with taxonomy, challenges, motivations, methodological aspects, recommendations, and pathways for future research. *Energy Strategy Reviews*, 45(September 2022), 101013. doi: [10.1016/j.esr.2022.101013](https://doi.org/10.1016/j.esr.2022.101013).
- Harshith Babu, V.S., Kumar, V., & Venkatesh, B.R. (2020). Consumer awareness and consumer perception towards green buying decisions an empirical study in Bangalore. *International Journal of Recent Technology and Engineering (IJRTE)*, 9(4), 8–12. doi: [10.35940/ijrte.d4740.119420](https://doi.org/10.35940/ijrte.d4740.119420).
- Ishak, S., & Zabil, N. F. M. (2012). Impact of consumer awareness and knowledge to consumer effective behavior. *Asian Social Science*, 8(13), 108–114. doi: [10.5539/ass.v8n13p108](https://doi.org/10.5539/ass.v8n13p108).
- Jain, P., & Hudnurkar, D. M. (2022). Sustainable packaging in the FMCG industry. *Cleaner and Responsible Consumption*, 7(November 2021), 100075. doi: [10.1016/j.clrc.2022.100075](https://doi.org/10.1016/j.clrc.2022.100075).
- Jha, R. K. (2023). Sustainable energy transition: Analyzing the impact of renewable energy sources on global power generation. October. doi: [10.36548/jaicn.2023.3.007](https://doi.org/10.36548/jaicn.2023.3.007).
- Kesari, K. K., Soni, R., Jamal, Q. M. S., Tripathi, P., Lal, J. A., Jha, N. K., . . . Ruokolainen, J. (2021). Wastewater treatment and reuse: A review of its applications and health implications. *Water, Air, and Soil Pollution*, 232(5), 208. doi: [10.1007/s11270-021-05154-8](https://doi.org/10.1007/s11270-021-05154-8).
- Keskin, B., Altay, B. N., Kurt, A., & Fleming, P. D. (2020). Sustainability in paper based packaging Kağıt Esashi Ambalajlarda Sürdürülebilirlik sustainability in paper based packaging. August.
- Khan, M., Farooq, N., Khattak, A., Hussain, A., Sahibzada, S., & Malik, S. (2020). Applying interpretive structural modeling and micmac analysis to evaluate inhibitors to transparency in humanitarian logistics. *Utopia y Praxis Latinoamericana*, 25(Extra2), 325–337. doi: [10.5281/zenodo.3815216](https://doi.org/10.5281/zenodo.3815216).
- Kitchen, P. J., & Vel, H. Y. B. S. P. (2005). The impact of implementing green supply chain management practices on corporate performance. *Competitiveness Review: An International Business Journal*, 15(1), 72–80. doi: [10.1108/10595420510818713](https://doi.org/10.1108/10595420510818713).

- Kukreti, A., Painolli, A. K., & Rana, N. (2021). Where do we stand: Factors affecting sustainable development. *Procedia Environmental Science, Engineering and Management*, 8(2), 583–589.
- Kumar, A. (2015). Green logistics for sustainable development: An analytical review. *IOSRD International Journal of Business*, 1(1), 7–13. Available from: www.iosrd.org
- Kumar, R., & Bangwal, D. (2023). An assessment of sustainable supply chain initiatives in Indian automobile industry using PPS method. *Environment, Development and Sustainability*, 25(9), 9703–9729. doi: [10.1007/s10668-022-02456-7](https://doi.org/10.1007/s10668-022-02456-7).
- Kumar, R., Agrawal, R., & Sharma, V. (2015). IT enablement in sugar supply chain: An approach for farmers. *International Journal of Business Performance and Supply Chain Modelling*, 7(4), 360–381. doi: [10.1504/IJBPSCM.2015.073770](https://doi.org/10.1504/IJBPSCM.2015.073770).
- Kumar, S., Smith, S. R., Fowler, G., Velis, C., Kumar, S. J., Arya, S., . . . Cheeseman, C. (2017). Challenges and opportunities associated with waste management in India. *Royal Society Open Science*, 4(3), 160764. doi: [10.1098/rsos.160764](https://doi.org/10.1098/rsos.160764).
- Lee, J., & Joo, H. Y. (2020). The impact of top management's support on the collaboration of green supply chain participants and environmental performance. *Sustainability (Switzerland)*, 12(21), 1–20. doi: [10.3390/su12219090](https://doi.org/10.3390/su12219090).
- Lee, H. T., Song, J. H., Min, S. H., Lee, H. S., Song, K. Y., Chu, C. N., & Ahn, S. H. (2019). Research trends in sustainable manufacturing: A review and future perspective based on research databases. *International Journal of Precision Engineering and Manufacturing - Green Technology*, 6(4), 809–819. doi: [10.1007/s40684-019-00113-5](https://doi.org/10.1007/s40684-019-00113-5).
- Lesic, V., De Bruin, W. B., Davis, M. C., Krishnamurti, T., & Azevedo, I. M. L. (2018). Consumers' perceptions of energy use and energy savings: A literature review. *Environmental Research Letters*, 13(3), 033004. doi: [10.1088/1748-9326/aaab92](https://doi.org/10.1088/1748-9326/aaab92).
- Li, X., Sohail, S., Majeed, M. T., & Ahmad, W. (2021). Green logistics, economic growth, and environmental quality: Evidence from one belt and road initiative economies. *Environmental Science and Pollution Research*, 28(24), 30664–30674. doi: [10.1007/s11356-021-12839-4](https://doi.org/10.1007/s11356-021-12839-4).
- Liu, T., Chen, L., Yang, M., Sandanayake, M., Miao, P., Shi, Y., & Yap, P. S. (2022). Sustainability considerations of green buildings: A detailed overview on current advancements and future considerations. *Sustainability (Switzerland)*, 14(21), 1–23. doi: [10.3390/su142114393](https://doi.org/10.3390/su142114393).
- Lukman, R. K., Glavič, P., Carpenter, A., & Vrtič, P. (2016). Sustainable consumption and production – research, experience, and development – the Europe we want. *Journal of Cleaner Production*, 138(December), 139–147. doi: [10.1016/j.jclepro.2016.08.049](https://doi.org/10.1016/j.jclepro.2016.08.049).
- Lv, Z., & Zhang, X. (2023). Influencing factor analysis on energy-saving refrigerator purchases from the supply and demand sides. *Sustainability (Switzerland)*, 15(13), 1–16. doi: [10.3390/su15139917](https://doi.org/10.3390/su15139917).
- Machado, C. G., Winroth, M. P., & Ribeiro da Silva, E. H. D. (2020). Sustainable manufacturing in industry 4.0: An emerging research agenda. *International Journal of Production Research*, 58(5), 1462–1484. doi: [10.1080/00207543.2019.1652777](https://doi.org/10.1080/00207543.2019.1652777).
- Madawaki, A. (2014). Impact of regulatory framework and environmental factors on accounting practices by firms in Nigeria. *Procedia - Social and Behavioral Sciences*, 164(August), 282–290. doi: [10.1016/j.sbspro.2014.11.078](https://doi.org/10.1016/j.sbspro.2014.11.078).
- Makanyeza, C., Sivotwa, T. D., & Jaiyeoba, O. (2021). The effect of consumer rights awareness on attitude and purchase intention in the hotel industry: Moderating role of demographic characteristics. *Cogent Business and Management*, 8(1). doi: [10.1080/23311975.2021.1898301](https://doi.org/10.1080/23311975.2021.1898301).
- Martín, J. C., Pagliara, F., & Román, C. (2019). The research topics on e-grocery: Trends and existing gaps. *Sustainability (Switzerland)*, 11(2), 1–15. doi: [10.3390/su11020321](https://doi.org/10.3390/su11020321).
- Mishra, B., & Kumar, A. (2023). How does regulatory framework impact sectoral performance? A systematic literature review. *International Journal of Productivity and Performance Management*, 72(5), 1419–1444. doi: [10.1108/IJPPM-07-2021-0398](https://doi.org/10.1108/IJPPM-07-2021-0398).
- Mitra, S. (2010). And reform in India background paper for OECD. 1–40. March.

- Mkanshi, M., Eresia-Eke, C., & Emmanuel-Ebikake, O. (2018). E-grocery challenges and remedies: Global market leaders perspective. *Cogent Business and Management*, 5(1), 1–28. doi: [10.1080/23311975.2018.1459338](https://doi.org/10.1080/23311975.2018.1459338).
- Montecchi, M., Plangger, K., & West, D. C. (2021). Supply chain transparency: A bibliometric review and research agenda. *International Journal of Production Economics*, 238(April), 108152. doi: [10.1016/j.ijpe.2021.108152](https://doi.org/10.1016/j.ijpe.2021.108152).
- Morashti, J. A., An, Y., & Jang, H. (2022). A systematic literature review of sustainable packaging in supply chain management. *Sustainability (Switzerland)*, 14(9), 4921. doi: [10.3390/su14094921](https://doi.org/10.3390/su14094921).
- Nie, H., Zhou, T., Lu, H., & Huang, S. (2021). Evaluation of the efficiency of Chinese energy-saving household appliance subsidy policy: An economic benefit perspective. *Energy Policy*, 149, 112059. doi: [10.1016/j.enpol.2020.112059](https://doi.org/10.1016/j.enpol.2020.112059).
- Nikolić, T. M., Paunović, I., Milovanović, M., Lozović, N., & Đurović, M. (2022). Examining generation Z's attitudes, behavior and awareness regarding eco-products: A Bayesian approach to confirmatory factor analysis. *Sustainability (Switzerland)*, 14(5), 2727. doi: [10.3390/su14052727](https://doi.org/10.3390/su14052727).
- Ofem, P., Isong, B., & Lugayizi, F. (2022). On the concept of transparency: A systematic literature review. *IEEE Access*, 10(January), 89887–89914. doi: [10.1109/ACCESS.2022.3200487](https://doi.org/10.1109/ACCESS.2022.3200487).
- Oloyede, O. O., & Lignou, S. (2021). Sustainable paper-based packaging: A consumer's perspective. *Foods*, 10(5), 1035. doi: [10.3390/foods10051035](https://doi.org/10.3390/foods10051035).
- Owusu, P. A., & Asumadu-Sarkodie, S. (2016). A review of renewable energy sources, sustainability issues and climate change mitigation. *Cogent Engineering*, 3(1), 1–14. doi: [10.1080/23311916.2016.1167990](https://doi.org/10.1080/23311916.2016.1167990).
- Perea-Moreno, A. J. (2021). Renewable energy and energy saving: Worldwide research trends. *Sustainability (Switzerland)*, 13(23), 11–13. doi: [10.3390/su132313261](https://doi.org/10.3390/su132313261).
- Prabowo, H., Hindarwati, E. N., & Yuniarty (2020). Online grocery shopping adoption: A systematic literature review. In *Proceedings of 2020 International Conference on Information Management and Technology*, August (pp. 40–45). ICIMTech. doi: [10.1109/ICIMTech50083.2020.9211241](https://doi.org/10.1109/ICIMTech50083.2020.9211241).
- Qazi, A., Hussain, F., Rahim, N. A. B. D., Hardaker, G., Alghazzawi, D., Shaban, K., & Haruna, K. (2019). Towards sustainable energy: A systematic review of renewable energy sources, technologies, and public opinions. *IEEE Access*, 7, 63837–63851. doi: [10.1109/ACCESS.2019.2906402](https://doi.org/10.1109/ACCESS.2019.2906402).
- Seroka-Stolka, O. (2014). The development of green logistics for implementation sustainable development strategy in companies. *Procedia - Social and Behavioral Sciences*, 151, 302–309. doi: [10.1016/j.sbspro.2014.10.028](https://doi.org/10.1016/j.sbspro.2014.10.028).
- Sharma, A. K., Nigrawal, A., & Baredar, P. (2020). Sustainable development by constructing green buildings in India: A review. *Materials Today: Proceedings*, 46(xxxx), 5329–5332. doi: [10.1016/j.matpr.2020.08.788](https://doi.org/10.1016/j.matpr.2020.08.788).
- Shree, S., Pratap, B., Saroy, R., & Dhal, S. (2021). Digital payments and consumer experience in India: A survey based empirical study. *Journal of Banking and Financial Technology*, 5(1), 1–20. doi: [10.1007/s42786-020-00024-z](https://doi.org/10.1007/s42786-020-00024-z).
- Siagian, H., Tarigan, Z. J. H., & Basana, S. R. (2022). The role of top management commitment in enhancing competitive advantage: The mediating role of green innovation, supplier, and customer integration. *Uncertain Supply Chain Management*, 10(2), 477–494. doi: [10.5267/j.uscm.2021.12.003](https://doi.org/10.5267/j.uscm.2021.12.003).
- Sikosana, M. L., Sikhwivhilo, K., Moutloali, R., & Madyira, D. M. (2019). Municipal wastewater treatment technologies: A review. *Procedia Manufacturing*, 35, 1018–1024. doi: [10.1016/j.promfg.2019.06.051](https://doi.org/10.1016/j.promfg.2019.06.051).
- Singh, A. P., & Rahman, Z. (2021). Integrating corporate sustainability and sustainable development goals: Towards a multi-stakeholder framework. *Cogent Business and Management*, 8(1). doi: [10.1080/23311975.2021.1985686](https://doi.org/10.1080/23311975.2021.1985686).
- Singh, A., Panackal, N., & Sharma, A. (2016). A study of environmental factors affecting industrial sustainability using ISM and MICMAC methodology. *International Journal of Applied Engineering Research*, 11(4), 2291–2297.

- Siracusa, V., & Rosa, M. D. (2018). Sustainable packaging. In *Sustainable Food Systems from Agriculture to Industry: Improving Production and Processing*, (November). doi: [10.1016/B978-0-12-811935-8.00008-1](https://doi.org/10.1016/B978-0-12-811935-8.00008-1).
- Siragusa, C., & Tumino, A. (2022). E-Grocery: Comparing the environmental impacts of the online and offline purchasing processes. *International Journal of Logistics Research and Applications*, 25(8), 1164–1190. doi: [10.1080/13675567.2021.1892041](https://doi.org/10.1080/13675567.2021.1892041).
- Stauropoulou, A., Sardonianou, E., Malindretos, G., Evangelinos, K., & Nikolaou, I. (2023). The role of customers' awareness towards the sustainable development goals (SDGs) of banks on their behavior. *Environmental Science and Pollution Research*, 30(5), 13495–13507. doi: [10.1007/s11356-022-23111-8](https://doi.org/10.1007/s11356-022-23111-8).
- Toke, L. K., & Kalpande, S. D. (2019). Critical success factors of green manufacturing for achieving sustainability in Indian context. *International Journal of Sustainable Engineering*, 12(6), 415–422. doi: [10.1080/19397038.2019.1660731](https://doi.org/10.1080/19397038.2019.1660731).
- Trivedi, P., & Sharma, M. (2017). Impact of green production and green technology on sustainability: Cases on companies in India. *International Journal of Mechanical and Production Engineering Research and Development*, 7(6), 591–606. doi: [10.24247/ijmperddc201767](https://doi.org/10.24247/ijmperddc201767).
- Turaga, R. M. R., & Sugathan, A. (2020). Environmental regulations in India. In *Oxford Research Encyclopedia of Environmental Science*, (August). doi: [10.1093/acrefore/9780199389414.013.417](https://doi.org/10.1093/acrefore/9780199389414.013.417).
- Tüzün Rad, S., & Gülmez, Y. S. (2017). Green logistics for sustainability. *International Journal of Management Economics and Business*, 13(3). doi: [10.17130/ijmneb.2017331327](https://doi.org/10.17130/ijmneb.2017331327).
- Varžinskis, V., & Markevičiūtė, Z. (2020). Sustainable food packaging: Materials and waste management solutions. *Environmental Research, Engineering and Management*, 76(3), 154–164. doi: [10.5755/j01.erem.76.3.27511](https://doi.org/10.5755/j01.erem.76.3.27511).
- Wang, H., & Liu, Z. (2019). The impact of top management commitment on companies' performance in green supply chain management. *Explicator*, 12(2), 1–15.
- Wang, C., Ghadimi, P., Lim, M. K., & Tseng, M. L. (2019). A literature review of sustainable consumption and production: A comparative analysis in developed and developing economies. *Journal of Cleaner Production*, 206, 741–754. doi: [10.1016/j.jclepro.2018.09.172](https://doi.org/10.1016/j.jclepro.2018.09.172).
- Yuan, X., Wang, X., & Zuo, J. (2013). Renewable energy in buildings in China-a review. *Renewable and Sustainable Energy Reviews*, 24, 1–8. doi: [10.1016/j.rser.2013.03.022](https://doi.org/10.1016/j.rser.2013.03.022).
- Yusliza, M. Y., Norazmi, N. A., Jabbour, C. J. C., Fernando, Y., Fawehinmi, O., & Seles, B. M. R. P. (2019). Top management commitment, corporate social responsibility and green human resource management: A Malaysian study. *Benchmarking*, 26(6), 2051–2078. doi: [10.1108/BJJ-09-2018-0283](https://doi.org/10.1108/BJJ-09-2018-0283).
- Zarte, M., Pechmann, A., & Nunes, I. L. (2022). Problems, needs, and challenges of a sustainability-based production planning. *Sustainability (Switzerland)*, 14(7), 4092. doi: [10.3390/su14074092](https://doi.org/10.3390/su14074092).

Further reading

- Singh, S. K., & Roy, S. (2020). Greening the grey: Measuring adoption of green logistics in Indian logistics sector. *International Journal of Logistics Systems and Management*, 1(1), 1. doi: [10.1504/ijlsm.2020.10031815](https://doi.org/10.1504/ijlsm.2020.10031815).

Corresponding author

Nida can be contacted at: nida.official01@gmail.com